

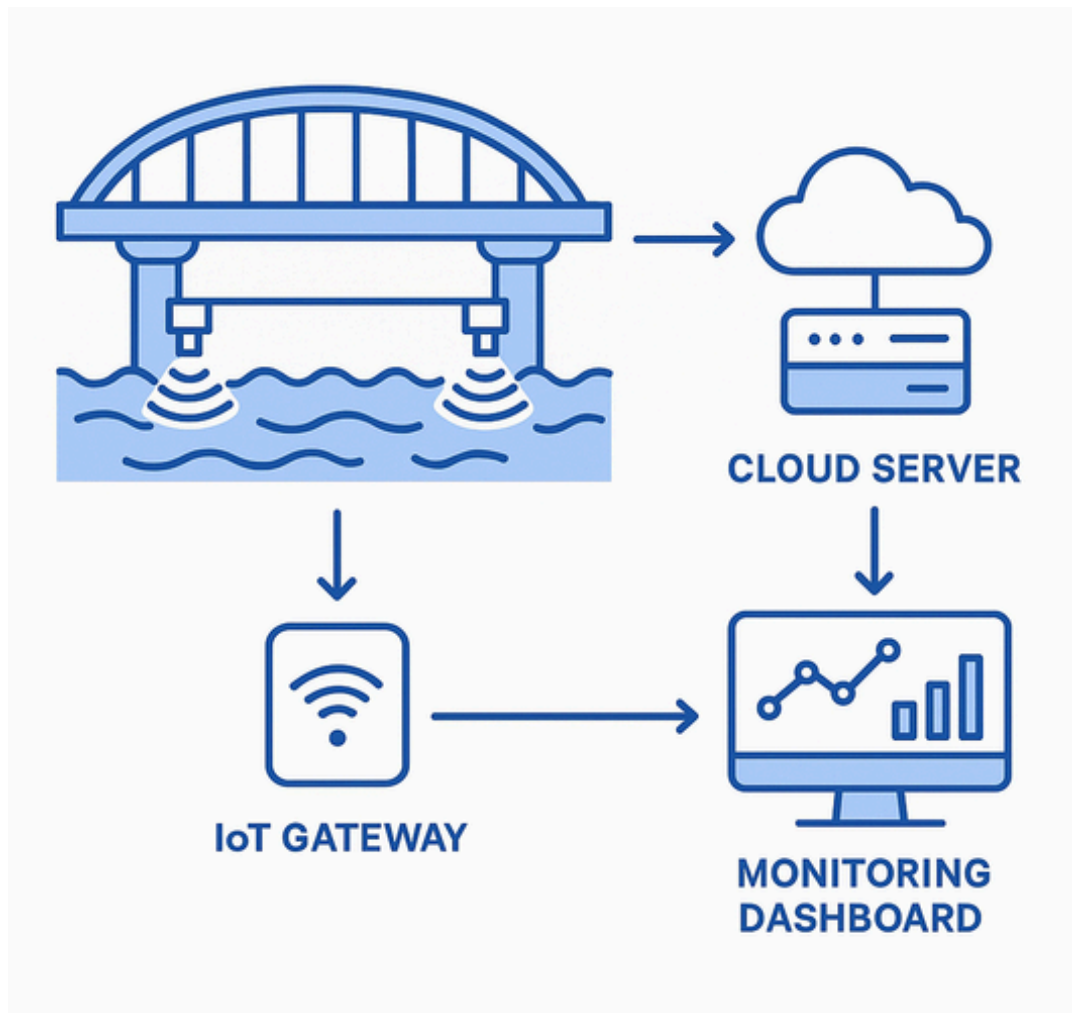


“Real-Time Scour Detection for Stronger Foundations”



Orbit Engineering’s Scour Monitoring System is designed to ensure the safety of rivers, bridges, and critical hydraulic structures. The system continuously monitors scour (erosion of soil around foundations) and provides real-time data so that preventive measures can be taken on time.

System Architecture



System Components

1. Radar Level Sensor

- Accurately measures water level and scour depth.
- Works on non-contact radar technology for reliable performance in all weather conditions.

2. PLC Control

- *Panel Acts as the brain of the system.*
- *Processes data from sensors. Logs temperature data and scour depth.*

3. Solar Power Supply

- *The site is completely remote with no external power source.*
- *The system is solar-operated with battery backup for uninterrupted operation.*



4. Cloud Connectivity

- *Data is transmitted 24/7 to the cloud.*
- *Enables remote monitoring from anywhere.*

3. Solar Power Supply

- *The site is completely remote with no external power source.*
- *The system is solar-operated with battery backup for uninterrupted operation.*

6. SMS/Report Alerts

- *Automated daily reports are sent to registered mobile numbers.*
- *Instant SMS alerts in case of abnormal conditions (high scour depth, unusual water level)*

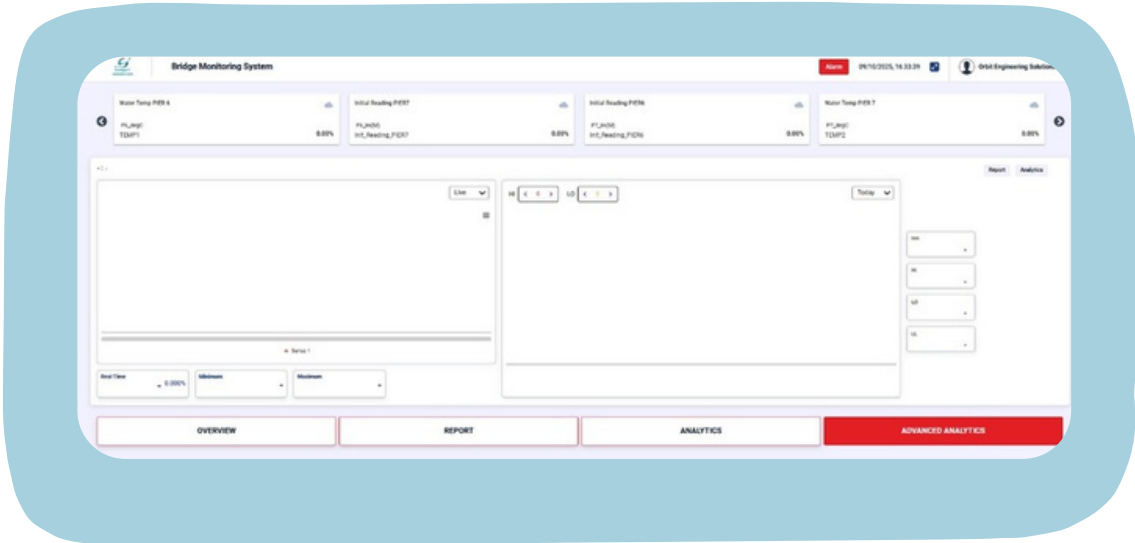
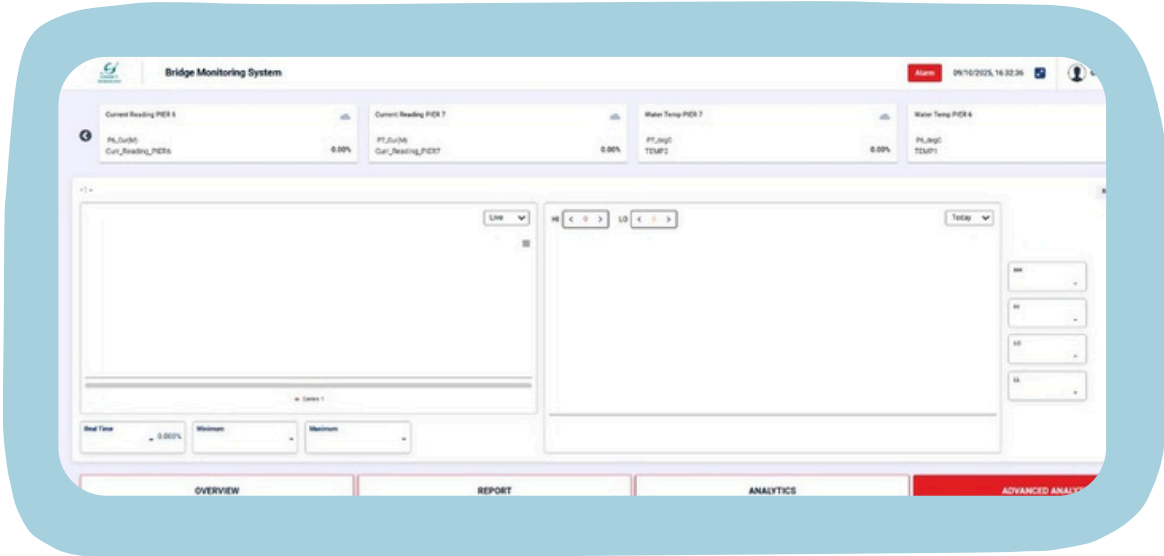
7. Electrical Waterproof Cables

- *PVC/XLPE Insulated Armored Cables – Basic waterproofing, used in general outdoor monitoring.*

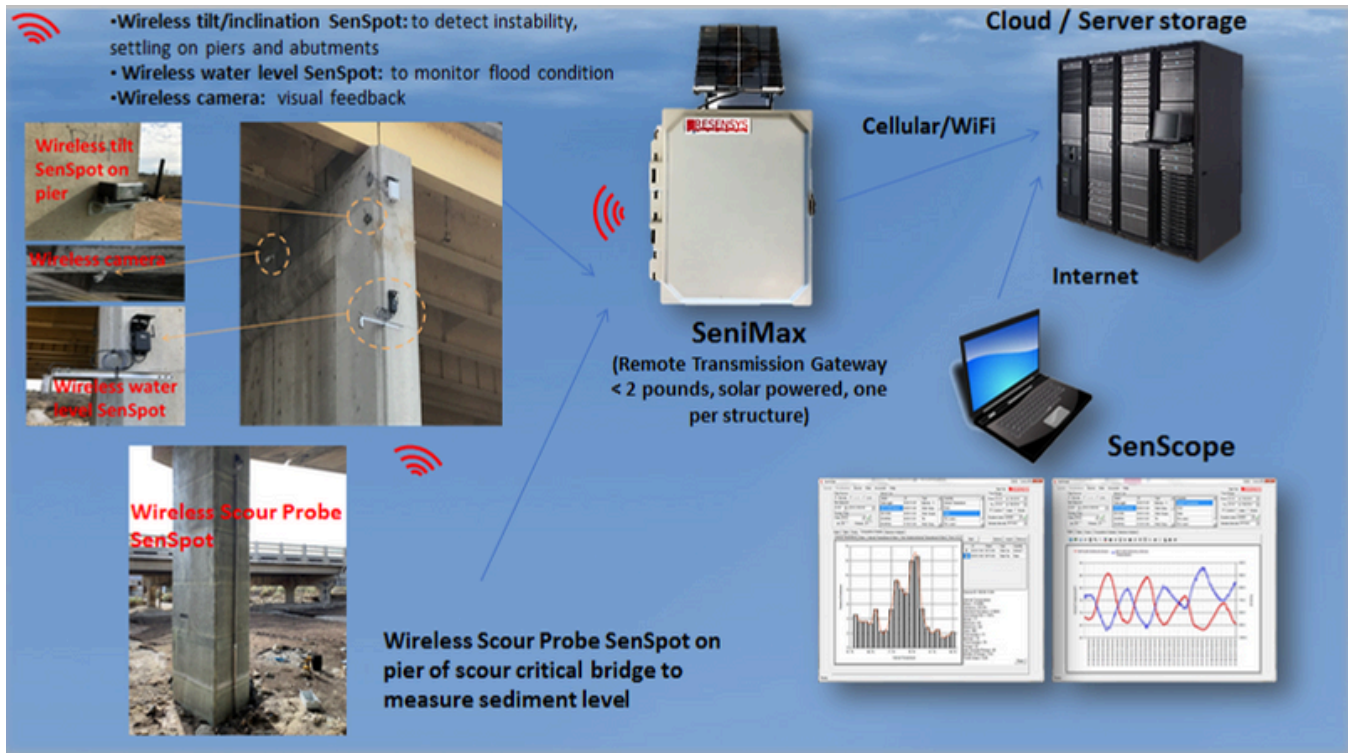
Site Activities



Cloud Login UI/UX



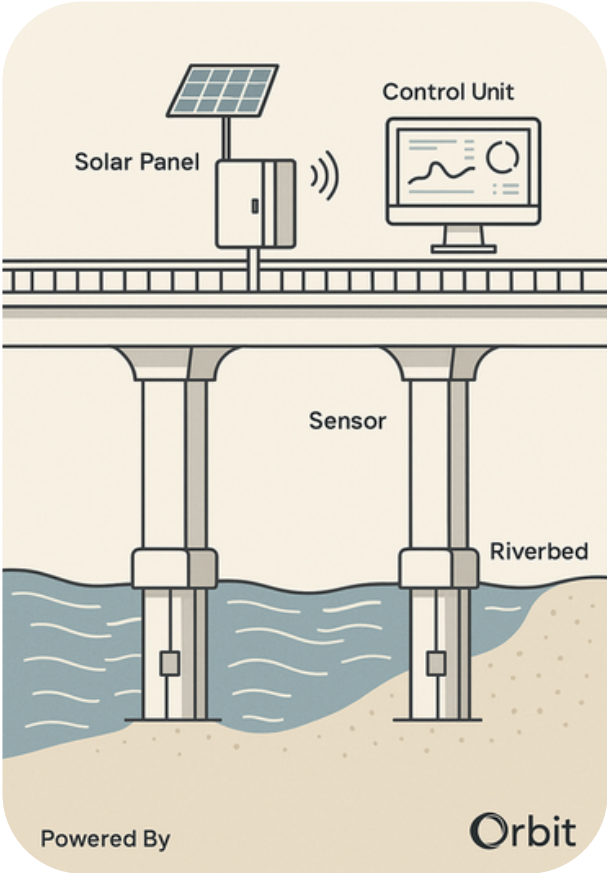
“Wireless Bridge Health and Scour Monitoring System”



“This system uses wireless SenSpot sensors to monitor bridge health and safety. It detects tilt, inclination, water level, and sediment scouring, sending real-time data to the cloud via a solar-powered gateway. Engineers can analyze the data through SenScope for preventive action.”

Scour Monitoring Panel

Follwing Images are Only For Refrance Purpose



Working Principle

- 1.Sonar Level Sensor continuously measures water level and scour depth.**
- 2.Data is transmitted to the PLC panel, where it is processed and stored.**
- 3.The solar-powered communication module sends data to the cloud.**
- 4.The cloud system visualizes this data on the UI/UX dashboard.**
- 5.D aily and periodic reports are automatically generated and shared with stakeholders via SMS.**

Key Feature

- Real-time scour & water level monitoring**
- Solar operated – independent of external power**
- PLC-based automation with local data logging**
- Cloud integration with remote monitoring**
- Interactive dashboard (UI/UX)**
- Automated daily SMS/Reports**
- Fully automated & low maintenance system**

Technical Specification

PulseSense Sonar

- 1. The Sonar Smart EcoSensor is a compact and intelligent sensor designed to deliver reliable detection and monitoring across diverse environments. Using advanced radar technology, it operates effectively in challenging conditions such as rain, fog, dust, and low light—where conventional sensors often fail. Its durable build and low power consumption make it an ideal choice for smart city projects, industrial automation, and traffic management applications.**

Key Features

- High Accuracy: Ensures precise detection in all environments.*
- All-Weather Performance: Functions seamlessly in rain, fog, dust, and darkness.*
- Compact & Easy Integration: Small design suited for multiple applications.*
- Energy Efficient: Low power usage for eco-friendly operation.*
- Rugged Design: Built to withstand tough outdoor and industrial conditions.*
- Smart Connectivity: Ready for IoT and modern automation platforms.*
- Versatile Applications: Perfect for vehicle detection, traffic monitoring, industrial safety, and smart infrastructure*



Report Generations

Report Name: Orin Bridge Monitoring System				Report Description: Daily Report & pm		Department: Engineering				Report Created By: [User Name]						
Report Type: Hour Log				Email Schedule: Daily		Role: Engineer				TimeRange: 2025-05-21 20:00:00						
Date Range		Time Range		Interval	Selected Gateway	Selected Machine	Machine UID	Plant	Selected Device	Units	Asset Health Data					
From	To	From	To								Min	Max	Mean	Std Dev	All Time Min	All Time Max
2025-05-20	2025-05-21	20:00:00	20:00:00	1	Orin Bridge Monitoring	Orin Bridge Monitoring	P1087410050P0/0RUMC0	Bridge Monitoring System	Orin Bridge Monitoring	P1_Cur(M)	1.7	129.9	9.79	7.84	3.16	141.18
										P1_Avg	10.09	11.48	10.27	0.74	10.58	46
										P1_Acc(M)	1	1	1	0	1	1
										P2_Cur(M)	1.18	10.48	2.82	1.24	1.01	81.43
										P2_Avg	17.26	32.8	24.26	2.8	18.22	40.86
										P2_Acc(M)	1	1	1	0	1	1

Date		Time		Orin Bridge Monitoring					
Date	Time	P1_Cur(M)		P1_Avg		P1_Acc(M)		P2_Cur(M)	
		P1_Cur(M)	P1_Avg	P1_Acc(M)	P1_Avg	P1_Acc(M)	P1_Avg	P2_Cur(M)	P2_Avg
2025-05-20	20:00:00	2.18	24.57	2.0	1.7	25.73	25.73	1.18	10.48
2025-05-20	20:01:00	2.18	24.57	2.0	1.79	25.44	25.44	1.18	10.48
2025-05-20	20:02:00	2.15	24.57	2.0	2.73	25.43	25.43	1.18	10.48
2025-05-20	20:03:00	2.49	24.55	2.0	2.88	25.58	25.58	1.18	10.48
2025-05-20	20:04:00	2.49	24.55	2.0	3.15	25.55	25.55	1.18	10.48
2025-05-20	20:05:00	2.43	24.48	2.0	3.3	25.55	25.55	1.18	10.48
2025-05-20	20:06:00	2.36	24.61	2.0	3.09	25.53	25.53	1.18	10.48
2025-05-20	20:07:00	2.47	24.59	2.0	3.27	25.46	25.46	1.18	10.48
2025-05-20	20:08:00	2.47	24.52	2.0	3.66	25.43	25.43	1.18	10.48
2025-05-20	20:09:00	2.7	24.3	2.0	3.85	25.37	25.37	1.18	10.48
2025-05-20	20:10:00	2.7	24.3	2.0	3.81	25.38	25.38	1.18	10.48
2025-05-20	20:11:00	2.4	24.3	2.0	2.86	25.38	25.38	1.18	10.48
2025-05-20	20:12:00	3.86	24.3	2.0	3.1	25.38	25.38	1.18	10.48
2025-05-20	20:13:00	3.3	24.3	2.0	2.73	25.38	25.38	1.18	10.48
2025-05-20	20:14:00	2.42	24.3	2.0	2.86	25.38	25.38	1.18	10.48
2025-05-20	20:15:00	2.7	24.37	2.0	2.86	25.38	25.38	1.18	10.48
2025-05-20	20:16:00	2.89	24.39	2.0	2.7	25.38	25.38	1.18	10.48
2025-05-20	20:17:00	2.89	24.38	2.0	2.86	25.38	25.38	1.18	10.48
2025-05-20	20:18:00	3.29	24.55	2.0	3.07	25.35	25.35	1.18	10.48
2025-05-20	20:19:00	3.77	24.55	2.0	2.85	25.39	25.39	1.18	10.48
2025-05-20	20:20:00	2.89	24.57	2.0	2.89	25.46	25.46	1.18	10.48
2025-05-20	20:21:00	2.55	24.55	2.0	1.9	25.49	25.49	1.18	10.48
2025-05-20	20:22:00	2.69	24.39	2.0	3.0	25.37	25.37	1.18	10.48
2025-05-20	20:23:00	2.55	24.3	2.0	2.88	25.38	25.38	1.18	10.48
2025-05-20	20:24:00	2.89	24.3	2.0	2.88	25.38	25.38	1.18	10.48
2025-05-20	20:25:00	1.26	24.3	2.0	2.79	25.37	25.37	1.18	10.48
2025-05-20	20:26:00	1.58	24.28	2.0	2.86	25.37	25.37	1.18	10.48
2025-05-20	20:27:00	2.77	24.37	2.0	2.85	25.23	25.23	1.18	10.48
2025-05-20	20:28:00	2.89	24.46	2.0	2.81	25.38	25.38	1.18	10.48
2025-05-20	20:29:00	1.88	24.55	2.0	2.84	25.38	25.38	1.18	10.48
2025-05-20	20:30:00	2.86	24.69	2.0	2.88	25.4	25.4	1.18	10.48
2025-05-20	20:31:00	1.55	24.48	2.0	2.81	25.35	25.35	1.18	10.48
2025-05-20	20:32:00	2.49	24.46	2.0	2.75	25.38	25.38	1.18	10.48
2025-05-20	20:33:00	3.3	24.46	2.0	2.85	25.38	25.38	1.18	10.48
2025-05-20	20:34:00	2.82	24.48	2.0	1.7	25.39	25.39	1.18	10.48
2025-05-20	20:35:00	3.2	24.3	2.0	2.88	25.32	25.32	1.18	10.48
2025-05-20	20:36:00	2.76	24.23	2.0	3.0	25.33	25.33	1.18	10.48
2025-05-20	20:37:00	2.89	24.23	2.0	2.7	25.37	25.37	1.18	10.48
2025-05-20	20:38:00	2.58	24.38	2.0	2.7	25.39	25.39	1.18	10.48
2025-05-20	20:39:00	2.09	24.37	2.0	2.55	25.36	25.36	1.18	10.48
2025-05-20	20:40:00	2.77	24.48	2.0	2.79	25.35	25.35	1.18	10.48
2025-05-20	20:41:00	2.89	24.3	2.0	2.85	25.37	25.37	1.18	10.48
2025-05-20	20:42:00	2.42	24.48	2.0	2.81	25.37	25.37	1.18	10.48
2025-05-20	20:43:00	2.77	24.55	2.0	2.84	25.37	25.37	1.18	10.48
2025-05-20	20:44:00	3.03	24.48	2.0	2.84	25.35	25.35	1.18	10.48
2025-05-20	20:45:00	2.82	24.48	2.0	2.55	25.38	25.38	1.18	10.48
2025-05-20	20:46:00	2.7	24.46	2.0	2.7	25.38	25.38	1.18	10.48
2025-05-20	20:47:00	2.77	24.39	2.0	2.9	25.35	25.35	1.18	10.48
2025-05-20	20:48:00	2.46	24.37	2.0	2.84	25.35	25.35	1.18	10.48
2025-05-20	20:49:00	2.77	24.3	2.0	2.75	25.3	25.3	1.18	10.48
2025-05-20	20:50:00	2.77	24.3	2.0	3.0	25.38	25.38	1.18	10.48
2025-05-20	20:51:00	2.86	24.3	2.0	2.86	25.38	25.38	1.18	10.48
2025-05-20	20:52:00	3.51	24.25	2.0	2.73	25.38	25.38	1.18	10.48
2025-05-20	20:53:00	3.04	24.23	2.0	2.81	25.23	25.23	1.18	10.48
2025-05-20	20:54:00	2.33	24.23	2.0	2.85	25.39	25.39	1.18	10.48
2025-05-20	20:55:00	2.89	24.38	2.0	3.16	25.36	25.36	1.18	10.48
2025-05-20	20:56:00	2.43	24.28	2.0	3.02	25.35	25.35	1.18	10.48
2025-05-20	20:57:00	2.77	24.3	2.0	2.83	25.36	25.36	1.18	10.48
2025-05-20	20:58:00	2.86	24.3	2.0	2.87	25.35	25.35	1.18	10.48
2025-05-20	20:59:00	2.89	24.62	2.0	2.88	25.39	25.39	1.18	10.48
2025-05-20	21:00:00	2.77	24.23	2.0	2.86	25.36	25.36	1.18	10.48
2025-05-20	21:01:00	2.77	24.23	2.0	3.09	25.44	25.44	1.18	10.48
2025-05-20	21:02:00	2.89	24.23	2.0	2.85	25.37	25.37	1.18	10.48
2025-05-20	21:03:00	2.77	24.12	2.0	2.82	25.38	25.38	1.18	10.48
2025-05-20	21:04:00	2.77	24.12	2.0	3.35	25.38	25.38	1.18	10.48
2025-05-20	21:05:00	2.89	24.07	2.0	2.84	25.38	25.38	1.18	10.48
2025-05-20	21:06:00	2.77	24.04	2.0	2.83	25.38	25.38	1.18	10.48
2025-05-20	21:07:00	2.77	24.06	2.0	2.82	25.39	25.39	1.18	10.48
2025-05-20	21:08:00	2.77	24.04	2.0	2.84	25.3	25.3	1.18	10.48
2025-05-20	21:09:00	2.77	24.04	2.0	3.15	25.35	25.35	1.18	10.48
2025-05-20	21:10:00	2.77	24.13	2.0	3.11	25.3	25.3	1.18	10.48